Ray **Garner**

Astronomy Postdoctoral Researcher, Texas A&M University

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Research Interests

- Nebular emission-line physics and diagnostics across cosmic time
- Chemical abundances and ionizing radiation in high-redshift galaxies with JWST
- Spatially resolved spectroscopy of H II regions and nearby galaxies
- Star formation, chemical evolution, spiral structure, and kinematics

Education

Case Western Reserve University — *Cleveland, OH*

Ph.D. in Astronomy — *Aug 2018 – Aug 2023*

- Thesis title: "Investigating the Galactic Evolution of M101 with Deep Narrowband Imaging"
- Committee: J. Christopher Mihos (Chair), Paul Harding, Steve A. Hauck, Bee Janesh, Stacy S. McGaugh
- Involves investigating the star formation, nebular abundance, and stellar populations of M101.

Furman University — *Greenville, SC*

B.S. in Physics, Summa Cum Laude — *Aug 2014 – May 2018*

Professional Experience

Texas A&M University, Physics & Astronomy Department — *College Station, TX*

Postdoctoral Researcher — *Sept 2023 – Present*

Casey Papovich & C3PO — *Sept 2026 – Present*

- Analyze JWST/NIRSpec rest-frame UV and optical spectroscopy of $6 < z < 9$ galaxies as part of the C3PO collaboration, with emphasis of nebular diagnostics, electron temperatures and densities, chemical abundances, and ionizing spectra.
- Develop Cloudy photoionization models to investigate hard ionizing photon production and abundance patterns in Reionization-era galaxies.
- Continue collaborative nearby-universe work through SIGNALS and ongoing CFHT/SITELLE and HETDEX projects.

Rob Kennicutt & SIGNALS — *Sept 2023 – Aug 2026*

- Led first-author studies of spatially resolved ionized gas physics as part of SIGNALS, a CFHT/SITELLE survey of approximately 50,000 H II regions across 35 nearby galaxies.
- Developed custom photoionization modeling and Bayesian inference methodologies for nebular abundances and ionization conditions.
- Obtained PI observing time with CFHT/SITELLE for high-resolution studies of NGC 604 and NGC 5474.
- Contributed to collaboration infrastructure and publication policy within an international research team.

Teaching — *2024 – Present*

- Co-taught ASTR-603/PHYS-643 *Stellar Astrophysics* in Fall 2024 and Fall 2026, with responsibility for approximately three weeks (5-6 lectures) of each semester.
- Guest and substitute lecturer for ASTR-320 *Astrophysical Research Methods*, ASTR-314 *Survey of Astronomy*, and ASTR-403 *Extragalactic Astronomy and Cosmology*.

Art of Problem Solving — *Online*

Instructor, Assistant, & Grader — *Aug 2015 – Feb 2016, May 2022 – Present*

- Promoted from Grader (Aug 2015–Feb 2016) to Assistant (May 2022–Sept 2023) to Instructor (Sept 2023–present), reflecting sustained excellence in teaching and student engagement.
- Teach and support online, live, text-based courses for advanced middle- and high-school students, fostering problem solving skills in highly motivative and mathematically talented learners.
- Instruct a broad portfolio of courses spanning physics, mathematics, and programming,
- Facilitate interactive, discussion-driven learning environments, guiding students through complex, multi-step problems and encouraging creative and rigorous approaches to problem solving.
- Implement AoPS's innovative, inquiry-based teaching methods, emphasizing deep conceptual understanding over rote memorization.

Case Western Reserve University, Astronomy Department *Cleveland, OH*

Teaching Assistant *Aug 2018 – Aug 2023*

- Supported instruction in undergraduate astronomy courses, including ASTR-101 *The Sun & Its Planets*, ASTR-105 *Introduction to Einstein's Universe*, ASTR-221 *Stars & Planets*, and ASTR-222 *Galaxies & Cosmology*.
- Led regular office hours, providing individualized academic support and mentoring to students across a range of experience levels.
- Delivered guest and substitute lectures in sophomore-level courses, presenting core concepts in stellar, planetary, and extragalactic astronomy.
- Graded assignments, offering detailed, constructive feedback to reinforce conceptual understanding and problem-solving skills.
- Organized and led Python help sessions each semester, introducing students to computational tools for data analysis and scientific problem solving.

Research Publications

Refereed Publications

Total refereed citations: 43 (first-author), 64 (co-author). h-index: 7

14. *H II Regions Filling Factor in NGC 628 and Spatial Association with Dusty Filaments*, Lorena Aragüete Riesco, José Manuel Vílchez, Salvador Duarte Puertas, …, **Ray Garner** (10th author), *Astronomy & Astrophysics*, 714, A233, 2026.
13. *SIGNALS of Giant H II Regions: A Spatially Resolved Analysis of NGC 604*, **Ray Garner**, Rob Kennicutt, Laurent Drissen, Carmelle Robert, Laurie Rousseau-Nepton, Christophe Morisset, Philippe Amram, R. Pierre Martin, & Emma Jarvis, *The Astrophysical Journal*, 997, 185, 2026.
12. *Optical Strong Line Ratios Cannot Distinguish Between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities*, Nikko J. Cleri, Grace M. Olivier, Bren E. Backhaus, …, **Ray Garner** (17th author), …, *The Astrophysical Journal*, 994, 146, 2025.
11. *Ionized gas in NGC 4258: Exploring the AGN - Star formation connection*, David Fernández-Arenas, Laurie Rousseau-Nepton, Carmelle Robert, …, **Ray Garner** (9th author), …, *Monthly Notices of the Royal Astronomical Society*, 542, 25, 2025.
10. *Peculiar Stellar Ages of the Peculiar Dwarf Galaxy NGC 5474*, Liv Moody & **Ray Garner**, *Research Notes of the AAS*, 9, 214, 2025.
9. *SIGNALS on the mixing of oxygen and nitrogen in the spiral galaxy NGC 6946*, Fabio Bresolin, David Fernández-Arenas, Laurie Rousseau-Nepton, **Ray Garner**, …, *Monthly Notices of the Royal Astronomical Society*, 539, 755, 2025.
8. *When Is a Bulge Not a Bulge? Revealing the Satellite Nature of NGC 5474's Bulge*, **Ray Garner**, J. Christopher Mihos, F. Fabián Rosales-Ortega, *The Astrophysical Journal*, 982, 143, 2025.
7. *NGC 628 in SIGNALS: Explaining the Abundance-Ionization Correlation in H II Regions*, **Ray Garner**, Rob Kennicutt, Laurie Rousseau-Nepton, Grace Olivier, David Fernández-Arenas, Carmelle Robert, Pierre Martin, & Philippe Amram, *The Astrophysical Journal*, 978, 70, 2025.
6. *Implications on star-formation-rate indicators from H II regions and diffuse ionized gas in the M101 Group*, Aaron E. Watkins, J. Christopher Mihos, Paul Harding, & **Ray Garner**, *Monthly Notices of the Royal Astronomical Society*, 530, 4560, 2024.
5. *A Dynamic Galaxy: Stellar Age Patterns across the Disk of M101*, **Ray Garner**, J. Christopher Mihos, Paul Harding, & Charles R. Garner, Jr., *The Astrophysical Journal*, 961, 217, 2024
4. *Spectroscopic Follow-up of Two Star-forming Objects in the M101 Field*, **Ray Garner**, Steven Janowiecki, & J. Christopher Mihos, *Research Notes of the AAS*, 7, 149, 2023.
3. *Deep Narrowband Photometry of the M101 Group: Strong-Line Abundances of 720 H II Regions*, **Ray Garner**, J. Christopher Mihos, Paul Harding, Aaron E. Watkins, & Stacy S. McGaugh, *The Astrophysical Journal*, 491, 182, 2022
2. *Determining the Timescale over Which Stellar Feedback Drives Turbulence in the Interstellar Medium: A Study of Four Nearby Dwarf Irregular Galaxies*, Laura Congreve Hunter, Liese van Zee, Kristen B. W. McQuinn, **Ray Garner**, and Andrew E. Dolphin, *The Astronomical Journal*, 163, 132, 2022
1. *A Deep Census of Outlying Star Formation in the M101 Group*, **Ray Garner**, J. Christopher Mihos, Paul Harding, and Aaron E. Watkins, *The Astrophysical Journal*, 915, 57, 2021

Submitted Manuscripts

3. *The N/O-O/H Relation Across Different Large-Scale Environments*, Yllari K. González-Koda, Laura Sánchez-Menguiano, Almudena Zurita, **Ray Garner** (4th author), …, *Astronomy & Astrophysics*, submitted.

2. *Mapping the Stellar Fossil Record in M101 with HETDEX: Spatially Resolved Ages and Metallicities*, **Ray Garner**, J. Christopher Mihos, Robin Ciardullo, Gregory Zeimann, Daniel J. Farrow, Olivia Curtis, Erin Mentuch Cooper, Caryl Gronwall, Gary J. Hill, Karl Gebhardt, Wolfram Kollatschny, & John J. Feldmeier, *The Astrophysical Journal*, submitted.
1. *Implications of Broad [O III]λ4364 and UV Line Emission in Two Little Red Dots at* $z \approx 7 - 8$, Casey Papovich, Kaila Ronayne, Weida Hu, …, **Ray Garner** (12th author), …, *The Astrophysical Journal*, submitted, arXiv:2608.04101.

Manuscripts in Preparation

4. *No AGN Required: Updated Photoionization Models for Rest-UV Line Diagnostics in Reionization-era Galaxies*, Jiayang Yang, Casey Papovich, Anthony J. Taylor, **Ray Garner**, …, in preparation.
3. *JWST TEMPLATES Resolved Metallicity Map for J1723+34*, Grace Olivier, Brian Welch, Taylor A. Hutchison, …, **Ray Garner** (8th author), …, *The Astrophysical Journal*, in preparation.
2. *H II Regions in M33 with SIGNALS: Identification and Physical Properties of Star-Forming Regions*, Emma Jarvis, Laurie Rousseau-Nepton, Carmelle Robert, Laurent Drissen, David Fernández-Arenas, **Ray Garner** (6th author), …, *The Astrophysical Journal*, in preparation.
1. *SHIINE: A PHANGS-MUSE Catalog of H II Region Internal Structure Across Nearby Galaxies*, Lorena Aragüete Riesco, Salvador Duarte Puertas, José Manuel Vílchez, Christophe Morriset, & **Ray Garner**, *Astronomy & Astrophysics*, in preparation.

Observing Programs

Canada-France-Hawaii Telescope, 3.6m

Instrument: SITELLE, a wide-field imaging Fourier transform spectrograph

- Awarded 10.1 hours in Semester 2026A for "The Ionized Gas of NGC 5474: Unravelling the Chemodynamics of a Peculiar Dwarf Galaxy with SITELLE."
- Awarded 4 hours DDT in Semester 2025B for "The Gas Kinematics of NGC 604: Kinematic Structure and Turbulence at High Resolution with SITELLE."

Hobby-Eberly Telescope, 10m

Instrument: Low Resolution Spectrograph 2

Awarded 1.83 hours DDT in Trimester 2023B for "Probing the Origin of Isolated Star-Forming Regions in the M101 Field." Resulted in a *Research Note*, Garner et al. (2023).

Research Supervision

Annabella Yang — *College Station, TX*

Topic: High-z C/O Abundances and Photoionization Modeling — *Sept 2026 – Present*

PhD student at Texas A&M University

Jelle Vandersnickt — *Leuven, Belgium*

Topic: Comparison of MUSE and SITELLE Data — *Jan 2025 – Present*

PhD student at KU Leuven, collaborator in SIGNALS

Liv Moody — *College Station, TX*

Topic: Stellar Populations of NGC 5474 — *Sept 2024 – Aug 2025*

Undergraduate student at Texas A&M University, graduated in Fall 2025. Culminated in a *Research Note* detailing our research.

Honors & Awards

2026	**Nominee**, 2026 President's Meritorious Staff Service Awards	*College Station, TX*
2025	**Certified**, Postdoc Success Program	*College Station, TX*
2025	**Certified**, Center for the Improvement of Mentored Experiences in Research (CIMER)	*College Station, TX*
2023	**Winner**, Lenore A. Kola Graduate Student Community Service Award	*Cleveland, OH*
2023	**Winner**, Graduate Dean's Instructional Excellence Award	*Cleveland, OH*

Leadership & Professional Service

Texas A&M University Postdoctoral Association

College Station, TX

Communications Officer (Jan 2024–present) and President (Aug 2025–June 2026) *Jan 2024 – Present*

- Elected to leadership of a postdoctoral association representing 650+ postdocs across Texas A&M University.
- As President, lead monthly Executive Committee meetings, set organizational strategy, and coordinate initiatives supporting postdoctoral professional development and community building.
- Represent postdoctoral interests in university governance, including the Council of Principal Investigators and University Staff Council, advocating for policy improvements and resource allocations.
- Partner with the Office of Postdoctoral Affairs to design and implement professional development programming and networking events.
- Selected to represent Texas A&M at the National Postdoctoral Association Annual Conference in 2026, engaging in national-level discussions on postdoctoral advocacy and policy.
- As Communication Officer, directed all organizational communications, including email and social media platforms, increasing visibility and engagement across the postdoctoral community.
- Co-organized the 8th, 9th, and 10th Annual Postdoctoral Research Symposium, coordinating across multiple postdoctoral groups and administrative units to support large-scale research showcases.

National Postdoctoral Association

College Station, TX

Co-Chair of Awards Subcommittee *June 2026 – Present*

- Selected as the Co-Chair of the Awards Subcommittee of the Meetings Committee of the NPA.
- Reviewed and set the criteria as well as selected the recipients of the travel awards.
- More broadly volunteered to help plan the Annual Conference of the National Postdoctoral Association.

Texas A&M University Astronomy Postdoc Group

College Station, TX

President (Sept 2024–May 2025) and Member (Sept 2023–present) *Sept 2023 – Present*

- Elected President of the Astronomy Postdoc Group; led monthly meetings and coordinated initiatives to support postdoctoral well-being, professional development, and community building within the department.
- Represented astronomy postdocs in faculty discussions, advocating for resources, improved communication, and a supportve research environment.
- Organized and hosted visiting speakers, facilitating scientific exchange and engagement within the department.
- Chaired sessions for the annual departmental AstroSymposium, supporting the presentation and dissemination of student and postdoctoral research.

Graduate Council of Arts & Sciences

Cleveland, OH

Treasurer (2020–2021) and Co-President (2021–2023) *Aug 2020 – May 2023*

- Elected to executive leadership of the Graduate Council of Arts & Sciences (GCAS), representing graduate students across the College of Arts & Sciences.
- Founded and implemented the GCAS Graduate Student Scholarship and Creative Endeavors Grants, securing funding to support five graduate students ($1000 each) for research and creative projects.
- Reviewed and evaluated grant applications as a member of the selection panel, contributing to equitable and merit-based funding decisions.
- Represented GCAS in the President's Roundtable, collaborating with university-wide graduate student leadership on advocacy and cross-campus initiatives.

Outreach & Public Engagement

Adopt-a-Physicist

Online

Volunteer *Oct 2018 – Present*

- Participate in a national outreach program connecting high school students with professional physicists for mentorship and career exploration.
- Engaged with students in an online forum, answering questions on physics concepts, academic pathways, and careers in STEM.
- Provided guidance on college preparation, research opportunities, and navigating scientific careers, supporting students from diverse backgrounds.

NASA/JPL Solar System Ambassadors

Various Locations

Ambassador

Jan 2017 – Present

- Selected as a NASA/JPL Solar System Ambassador, a national outreach program promoting public engagement with NASA missions and discoveries.
- Delivered 35+ public presentations at libraries, schools, and community events nationwide, reaching diverse audiences of students and the general public.
- Developed and presented engaging talks on planetary exploration, the night sky, and current space missions, translating complex scientific concepts into accessible narratives.
- Fostered public interest in space science and increased awareness of NASA's scientific and exploratory efforts through community-based outreach.
- Adapted presentation style and content for audiences ranging from K-12 students to adult learners, emphasizing inclusive and interactive scientific communication.

Science Olympiad

Mentor, OH & College Station, TX

Coach & Test Writer

Sept 2021 – Present

- Coached the Mentor High School Division C Astronomy Team (2021-2023), leading weekly training sessions on advanced astronomy concepts and competition strategies.
- Authored and developed Division C Astronomy exams for multiple competitions, including the Mentor Invitational (2022-2026), Kenston Invitational (2023), and the Texas Science Olympiad Tournament (2024-2026).
- Proctored and administered Science Olympiad events, including Division B Reach for the Stars (Northeast Ohio Regional, 2022-2023) and Division C Astronomy (Texas State Tournament, 2025), ensuring fair and effective competition execution.

Presentations

Invited Talks & Colloquia

8. **AggieSTAAR: Introduction to H II Regions**, a lunch talk presented to undergraduates as part of the Aggie Scholarships for Technology Advancements in Astronomical Research (AggieSTAAR) program at Texas A&M University in College Station, TX, June 2025.
7. **The SIGNALS from H II Regions**, an invited review talk presented at the "SIGNALS and Friends" workshop at Université Laval in Québec City, QC, Canada, May 2025.
6. **SIGNALS of Star Formation in Local Galaxies**, presented as part of a seminar series at CWRU in Cleveland, OH, Feb 2025.
5. **Stellar Age Patterns Across the Disk of M101 with Narrowband Imaging**, presented virtually as part of a symposium series at the Centro de Estudios de Fisica del Cosmos de Aragón in Teruel, Spain, Dec 2023.
4. **Investigating the Galactic Evolution of M101 with Deep Narrowband Imaging**, presented as part of a colloquium series at the University of Toronto in Toronto, ON, Canada, Oct 2023.
3. **Galactic Interactions and Peculiar Galaxies**, presented as part of a colloquium series for the CWRU Undergraduate Research Society in Cleveland, OH, Mar 2023.
2. **Investigating the Oxygen Abundance of the M101 Group with Deep, Narrowband Imaging**, presented as part of a colloquium series at the Oberlin College & Conservatory in Oberlin, OH, Nov 2022.
1. **The M101 Interaction & Narrowband Photometry**, virtually to the LSST LSB Working Group, June 2022.

Conference Presentations

7. **Mapping the Stellar Fossil Record of M101 with HETDEX: Spatially Resolved Ages and Metallicities**, presented as a poster at the 10th Annual Postdoctoral Research Symposium at Texas A&M University in College Station, TX, Sept 2026.
6. **SIGNALS of Giant H II Regions: A Spatially Resolved Analysis of NGC 604**, presented in some form under various titles at:
    - the 247th Meeting of the AAS in Phoenix, AZ, Jan 2026.
    - the 9th Annual Postdoctoral Research Symposium at Texas A&M University in College Station, TX, Sept 2025.
5. **NGC 628 in SIGNALS: Explaining the Abundance-Ionization Correlation in H II Regions**, presented in some form under various titles at:
    - the "SIGNALS and Friends" workshop at Université Laval in Québec City, QC, Canada, May 2025.
    - the 8th Annual Postdoctoral Research Symposium at Texas A&M University in College Station, TX, Sept 2024.
    - the "Star Formation Across the Scales: Star Clusters to Galactic Disks" workshop at McMaster University in Hamilton, ON, Canada, Aug 2024.

    - the “Recipes to Regulate Star Formation at All Scales: From the Nearby Universe to the First Galaxies” symposium as a poster at the Space Telescope Science Institute in Baltimore, MD, Apr 2024.
    - the Joint Nuclear and Astrophysics Seminar series Texas A&M University in College Station, TX, Feb 2024.
4. **Investigating the Galactic Evolution of M101 with Deep Narrowband Imaging**, presented as my PhD thesis defense at CWRU in Cleveland, OH, June 2023.
3. **Investigating the M101 Group with Deep Narrowband Imaging**, presented as a 15-minute dissertation session at the 241st Meeting of the AAS in Seattle, WA, Jan 2023.
2. **Searching for Extragalactic Star Formation in the M101 Group**, virtual iPoster at the 237th Meeting of the AAS, Jan 2021.
1. **Ionized Gas Kinematics of Nearby Low-Mass Galaxies**, presented at
    - Furman Engaged at Furman University in Greenville, SC, Apr 2018.
    - 231st Meeting of the AAS in Washington, DC, Jan 2018.
    - SoCon Undergraduate Research Forum at Wofford College in Spartanburg, SC, Oct 2017.

Selected Outreach Talks

I list only the ten most recent talks. A more complete list can be found on my website here.

10. **How to be a Successful Postdoc**, invited talk presented at the 2026 Frontiers Conference at Texas A&M University in College Station, TX, May 2026.
9. **Astronomy on Tap: The Cursed History of Star Brightness**, presented at The 101 in Bryan, TX, Nov 2025.
8. **Astronaut Day**, an event for local high schoolers, presented at Texas A&M University in College Station, TX, Nov 2025.
7. **Astronomy on Tap: 110 Objects That Are Not Comets**, presented at The 101 in Bryan, TX, Feb 2025.
6. **Astronomy on Tap: Star-forming Regions and How to Read Them**, presented at the Grand Stafford Theater in Bryan, TX, Oct 2024.
5. **The Apollo Missions and the Moon Landing**, presented at the
    - Brazos Valley Astronomy Club in College Station, TX, Sept 2024.
    - Martz-Kohl Observatory in Frewsburg, NY, July 2022.
    - Andover Public Library in Andover, OH, July 2019.
4. **A Journey Through the Night Sky**, presented at the
    - Navasota Public Library in Navasota, TX, Aug 2024.
    - Brazos Valley Astronomy Club in College Station, TX, Jan 2024.
    - Andover/Pymatuning Lake KOA in Williamsfield, OH, July 2022.
    - Kent Free Library in Kent, OH, May 2022.
3. **2024 Solar Eclipse Project**, presented in collaboration with the Friends of Chamber Music and the Brazos Valley Astronomy Club in College Station, TX, Apr 2024.
2. **The 2024 Solar Eclipse**, presented at the Navasota Public Library in Navasota, TX, Apr 2024.
1. **An Annular Solar Eclipse in Texas!**, presented at the Navasota Public Library in Navasota, TX, Oct 2023.